Newton's laws of motion

NEWTON'S LAWS EXPLAIN HOW FORCES ACT ON OBJECTS.

When a force acts on an object that is free to move, the object will move in accordance with Newton's three laws of motion.

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A new direction for physics

Isaac Newton (1642–1727) published his laws of motion in 1687, setting the direction for physics over the next two centuries. He explained that when the forces acting on an object are balanced, there is no change in the way it moves. When the forces are unbalanced, there is an overall force in one direction, which alters the object's speed or the direction in which it is moving. Newton also emphasized the complicated relationship between objects and forces, which is due mainly to the effects of friction and air resistance. Without these effects, he concluded, the motions of objects are much simpler. So, his laws apply most obviously to bodies in space, such as planets and spacecraft.

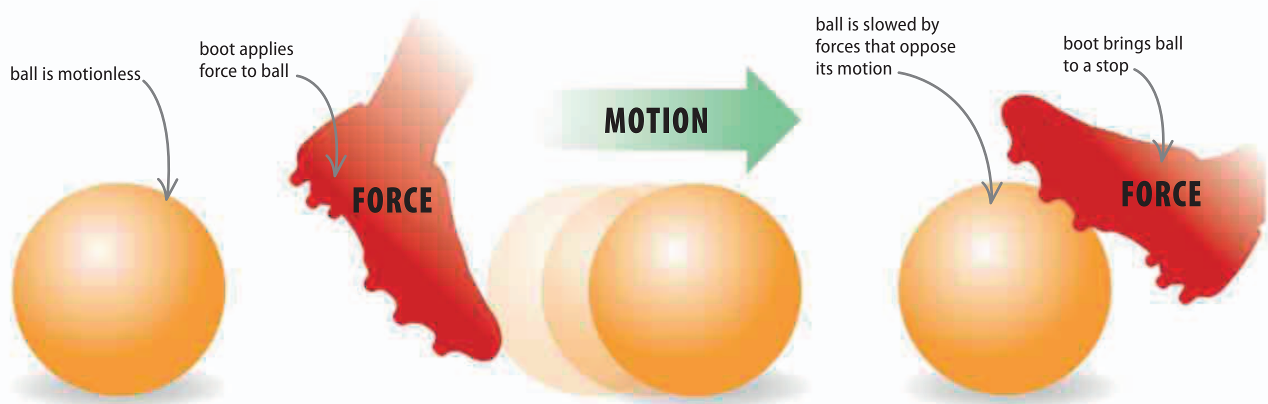
REAL WORLD

Blast off!

Newton's laws of motion can be used to explain how a rocket blasts into space. At the start, there is no force acting on the rocket, so it does not move. Then, when the rocket's engines fire, the force they produce lifts the rocket up and off the launchpad. As the hot gases shoot down, an equal force pushes the rocket up.

First law of motion

The first law of motion states that any object will continue to remain stationary, or move in a straight line at a constant speed, unless an external force acts on it. So, a soccer ball is stationary until it is kicked, and then it moves until other forces bring it to a halt.



△ **At rest**
Although the force of gravity is acting on the soccer ball, the ground below it stops the ball from moving, so it remains in a state of rest.

△ **Force applied**
The impact of a boot applies a force to the ball. For as long as the boot is in contact with the ball, the ball will be accelerated by it.

△ **Motion is arrested**
The ball immediately begins to slow due to the resistance of the air and friction with the ground. It is brought to a stop when it encounters a stationary object (the boot).